

Sentinel Prime AI — Anomaly Detection: Operational Framing

Week 11 · public-data prototype (NAB + SKAB) · no InGen data used. The question this answers: **what threshold should the product ship with, to keep recall above a stated bar without drowning operators in false alarms?**

Recommendation (headline)

Ship a sustained-anomaly rule, not a point rule.

On the best model (AutoEncoder), alert only after **24 consecutive readings** (120 minutes sustained) above score **2.16**. That catches **4/4 true failure events** at **88% point-recall** while raising only **0.50 false alerts per day** — versus **16.35/day** for the same threshold with no persistence filter.

Why the usual metric misleads

At an 80% recall floor the point-level false-alarm rate sits near **13%** whichever threshold or persistence setting we pick — it is fixed by how well the model separates the classes (AP=0.51), not by threshold policy. But an operator never experiences '13% of readings'; they experience **how often the phone buzzes**. Counting one alert per contiguous run — the way a product pages — a 120-minute persistence filter cuts alert load **33×** (16.35 → 0.50/day) while recall *risks* to 88%: the real failures are sustained, the false positives are isolated spikes.

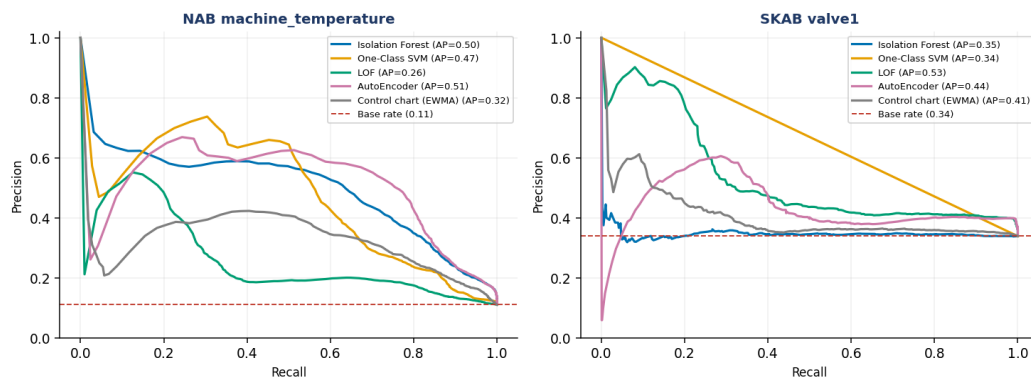
Stated bars, and an honest gap

Recall floor: 80% of anomalous readings. **False-alarm budget: 5%** of normal readings. **The point-level budget is not reachable** here: at 80% recall the achievable false-alarm rate bottoms out near 13%. Closing that needs a better model (richer context/sensor fusion), not more tuning. At the *alert* level the persistence rule already delivers a comfortable 0.5 false alerts/day.

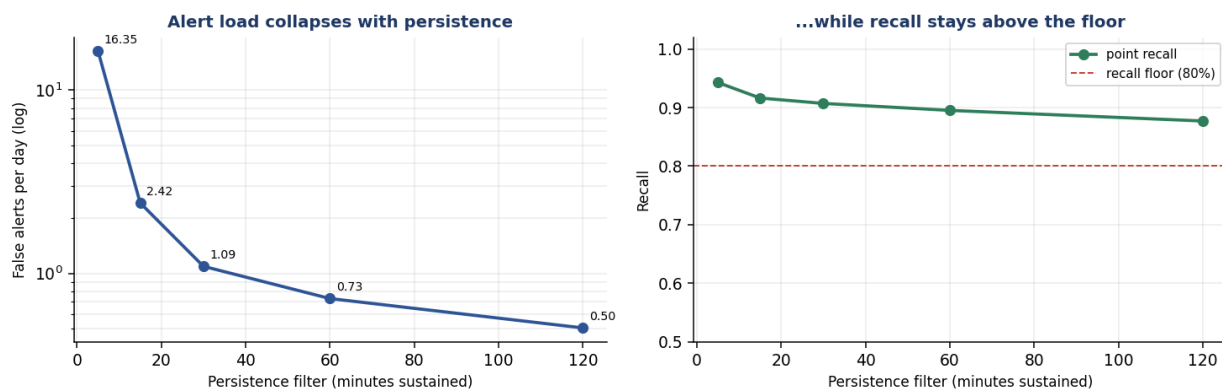
Model comparison — 5 detectors, identical splits

Dataset	Model	P	R	F1	AP	FAR	TTD (s)	Events
NAB	Isolation Forest	0.349	0.800	0.486	0.499	0.185	0	4/4
NAB	One-Class SVM	0.237	0.800	0.365	0.475	0.320	7125	4/4
NAB	LOF	0.174	0.800	0.286	0.261	0.469	20400	4/4
NAB	AutoEncoder	0.425	0.800	0.555	0.512	0.134	4800	4/4
NAB	Control chart (EWMA)	0.253	0.800	0.385	0.323	0.292	1875	4/4
SKAB	Isolation Forest	0.346	0.842	0.490	0.346	0.815	0	3/3
SKAB	One-Class SVM	0.339	1.000	0.506	0.339	1.000	0	3/3
SKAB	LOF	0.412	0.805	0.545	0.531	0.588	1	3/3
SKAB	AutoEncoder	0.403	0.846	0.546	0.436	0.640	31	3/3
SKAB	Control chart (EWMA)	0.360	0.800	0.496	0.405	0.729	0	3/3

AutoEncoder leads on NAB (F1 0.555, AP 0.512); LOF leads on SKAB (F1 0.545, AP 0.531). Both beat the EWMA control-chart baseline — the bar that justifies the added complexity. On SKAB every model shows a high false-alarm rate: that run is 34% anomalous, straining the 'mostly normal' assumption behind unsupervised detection, so NAB is the more realistic Sentinel proxy.



Threshold policy — fixed vs adaptive (and what failed)



Adaptive thresholds lost here — and the reason matters

Scheme	Precision	Recall	F1	FAR
Fixed threshold	0.425	0.800	0.555	0.134
Adaptive — rolling quantile (W=1000, q=0.98)	0.556	0.334	0.417	0.033
Adaptive — EWMA + 3.0sigma (alpha=0.01)	0.186	0.024	0.043	0.013
Adaptive — rolling quantile + baseline freeze (W=1000, q=0.98)	0.203	0.832	0.326	0.405

A rolling-quantile threshold cut false alarms hard (13.4% → 3.3%) but **collapsed recall (80% → 33%)**. Diagnosis: NAB's failure windows run ~2 days (~576 readings), so a baseline computed over recent history **absorbs the anomaly into itself** and lifts the threshold exactly when it should hold. This was not a tuning artefact — it survived a sweep of windows (1k–12k readings) and EWMA rates (α 0.01–0.0002). Freezing the baseline during an alert confirmed the diagnosis (recall recovered to 83%) but over-corrected: with the baseline frozen it stops tracking legitimate drift and false alarms jump to 40%.

Verdict: adaptive thresholding is the right tool when the drift timescale is much longer than the anomaly duration. For sustained failures like these, a fixed threshold calibrated on clean commissioning data plus a persistence filter is both simpler and better. Sentinel should default to fixed + persistence, and re-baseline on a schedule (or on a site change) rather than continuously.

Suggested product behaviour: two tiers

Tier	Rule	Purpose
Fast	score $\geq$ high cut-off, no persistence	Immediate page for unambiguous events (intrusion). Accepts more false alarms by design.
Sustained	score $\geq$ 2.16 for 120 min	Low-noise channel for degradation/failure. ~0.5 false alerts/day; TTD $\geq$ persistence window.

The persistence filter buys quiet at the cost of latency: a 120-minute rule cannot detect anything faster than 120 minutes. That is acceptable for slow degradation, and unacceptable for intrusion — hence two tiers rather than one compromise threshold.

Data & licences

Dataset	Series used	Licence
NAB https://github.com/numenta/NAB	realKnownCause/machine_temperature_system_failure.csv	MIT License (Copyright 2014-2024 Numenta Inc.)
SKAB https://github.com/waico/SKAB	valve1/0.csv, valve1/1.csv, valve1/2.csv	GNU General Public License v3 (GPL-3.0)

Both benchmarks are real public sensor data with published labels; no InGen data was used, per the Week 11 brief. Splits: NAB trains on all readings before the first labelled window (a clean warm-up, mirroring commissioning); SKAB trains on its shipped anomaly-free reference run. Detectors are unsupervised — labels are used only to evaluate and to select the operating point. Seeded (random_state=42) and reproducible via `python -m src.anomaly.run_all`.